

ZeroDrift Mk3 — buy checklist, one item per row

Total ₹6,709 · ✓ = price verified against a live listing, ~ = estimate · generated from docs/data/bom_tier_a.json

Tick the box for wherever you actually buy each item — a filled box is my routing guess, an empty one means I'd default to Amar Robotics unless it's clearly a specialist part. I never saw Amar Robotics' actual stock (Instagram/IndiaMart are both blocked from where I work), so treat every box as a suggestion to confirm, not a fact. The five **online** items are the ones with real delivery lead time — order those first, tonight, regardless of what else is still open.



DO NOT BUY THE OV7670. It is a parallel-DVP sensor, not a USB camera, and it cannot work here for three independent reasons. (1) No USB — the pipeline runs on the laptop and opens the camera with OpenCV; the OV7670 exposes D0-D7, PCLK, HREF, VSYNC and SCCB, which no laptop can read. (2) The Nano cannot read it either: ~12 MHz pixel clock against a 16 MHz ATmega328P with **2 KB** of SRAM, when one 640×480 frame is **307 KB**. (3) The bare board has no AL422B FIFO, and even the FIFO version manages a few FPS at 160×120 where the spec needs **≥20 FPS at 640×480**.



BUY THIS INSTEAD. Ask for a **USB webcam module, UVC, 640×480 or better, M12 lens**. The test is simple: plug it into a laptop, and if it appears as an ordinary webcam it will work.



THE MAGNET IS THE ONE THAT LOOKS IDENTICAL. A diametric magnet is polarised **across its diameter**; an axial magnet is polarised **through its thickness**. Same cylinder, same size, same price, same silver finish — and an axial magnet spinning in front of an AS5600 presents an **unchanging field**, so the encoder reads a constant and both feedback axes are dead. Nothing in the build tells you why. **Say the word “diametric” out loud at the counter and check the listing text before paying.**

QTY	ITEM		₹	AMAR ROBOTICS	BLINKIT	ONLINE	HAVE
	2	NEMA17 stepper motor, 1.8°/step, 4.2 kg-cm	✓	₹1,498	✓		
	2	A4988 stepper driver module	✓	₹340	✓		
	2	NEMA17 L-bracket (pan + tilt)	✓	₹254	✓		
	2	100 µF 25 V electrolytic capacitor	~	₹20	✓		
	2	AS5600 magnetic encoder module	✓	₹498		✓	
	2	Diametric magnet, 6 × 2.5 mm	~	₹180		✓	
	1	TCA9548A I²C multiplexer	✓	₹199		✓	
	1	USB webcam, UVC, 640x480+ — housing removed before mounting	✓	₹800		✓	
	1	Red filter — gel sheet or red acrylic offcut	~	₹100		✓	
	—	KY-008 laser module	~	₹0	✓		
	1	ABS project box, head enclosure	~	₹115	✓		
	—	Arduino Nano — tracker	~	₹0	✓		
	1	Arduino Nano — beacon	~	₹225	✓		
	1	12 V Li-ion pack, 3S (or 3x18650 + holder)	~	₹650		✓	
	1	3S 12.6 V balance charger — NOT a single-cell charger	~	₹250		✓	
	1	DC barrel jack to screw-terminal adapter	~	₹30	✓		
	1	SPST toggle switch — motor rail kill switch	~	₹25	✓		
	1	Breadboard, 830 point (drivers + Nano + mux)	~	₹90	✓		
	2	USB cables for the two Nanos — check each port before buying	~	₹198		✓	
	1	Acrylic sheet 3 mm, 6”×6” (pack of 2)	✓	₹199	✓		
	1	Vibration motor, 3 V DC	✓	₹20	✓		
	1	Transistor + diode + resistor combo kit	✓	₹105	✓		
	1	High-brightness RED LED, 10 mm, 650 nm	~	₹22	✓		

QTY	ITEM		₹	AMAR ROBOTICS	BLINKIT	ONLINE	HAVE
	1	2N2222 transistor + 220 Ω + 1 kΩ	~	₹20	✓		
	1	Ping-pong ball (diffuser)	✓	₹129		✓	
	—	USB power bank (any, 5V) — beacon supply	~	₹0	✓		
	1	SPST toggle switch	~	₹25	✓		
	1	ABS project box	~	₹115	✓		
	1	High-brightness WHITE LED (brighter than beacon)	~	₹22	✓		
	1	68 Ω resistor	~	₹5	✓		
	1	3×AA battery holder + 3 AA cells — decoy supply	~	₹90	✓		
	1	Switch + scrap housing	~	₹35	✓		
	1	Dupont jumper wires — M-M, M-F, F-F	~	₹150	✓		
	1	M3 screws, nuts, standoffs assortment	~	₹150	✓		
	1	Heat-shrink, solder, hot-glue sticks	~	₹150	✓		

GRAND TOTAL			₹6,709	₹3,705	₹327	₹2,677	₹0
Amar Robotics — local shop, phone ahead for the four motion parts			Blinkit — ping-pong ball only; a sports shop is ~4x cheaper if the 10 min don't matter		Online — Amazon/Robu/ThinkRobotics, order tonight		Have — already own it, nothing to buy

ZeroDrift · Team SIH26169 · ISRO · full guide in docs/submission/ZeroDrift_Mk3_Build_Guide.pdf · checklist generated from the same source file the guide uses, so the two cannot disagree

TOOLS — NOT PART OF THE ₹6,709, NOT THE BUILD

Not part of the build and not in the total. These are what you need in your hand to assemble it. Tick 'have' for anything the team or the college lab already owns -- most teams own most of this. The multimeter is the one that is genuinely not optional.

TOOL	WHY YOU NEED IT	₹ IF BUYING	HAVE	BUY
Multimeter	Setting the A4988 Vref is a voltage measurement on the driver's trim pot. Without it you are guessing the motor current: too low and the gimbal skips steps, too high and the motor cooks. There is no way to do this by eye.	₹500		
Small flat screwdriver, 2 mm	The A4988 trim pot is tiny and a normal screwdriver slips off it onto the board.	₹150		
Wire stripper / side cutter	Thirty-odd connections.	₹150		
Soldering iron, 25-40 W	The beacon and decoy are soldered; solder itself is already in consumables.	₹350		
Hot glue gun	Strain relief on every connector that leaves a moving axis. Glue sticks are already in consumables.	₹250		
Allen key set (M3)	NEMA17 brackets use M3 socket screws.	₹120		
Zip ties + double-sided tape	Cable management on a gimbal is not cosmetic: a cable that snags is a lost step and a lost lock.	₹80		
USB extension cable, 1.5 m	The camera rides on the gimbal; its cable has to reach the laptop with slack for the full pan range.	₹150		
IF YOU OWN NONE OF IT		₹1,750		

SCAN FOR REAL PHOTOS & TODAY'S PRICES — point your phone camera at any code



USB camera module
(UVC)



Diametric magnet
6x2.5



AS5600 encoder
board



A4988 driver



TCA9548A I2C mux



NEMA17 + L-bracket



12V Li-ion pack



Barrel to screw term.